

Improving Control Performance by Minimizing Jitter in RT-WiFi Networks

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ABSTRACT

While RT-WiFi provides guaranteed delivery latency for networked control systems, communication jitter (delay variation) significantly impacts control performance. The authors design an RT-WiFi network manager that generates data link layer communication schedules with minimal jitter, applicable to both static and dynamic network topologies. The approach uses a two-stage scheduling method and a novel S-tree data structure to efficiently handle network dynamics with minimal overhead, validated on a gait rehabilitation system.

RELEVANCE TO THE EB15 ROBOTIC ARM PROJECT

Provides the theoretical foundation for minimizing communication latency and jitter in the wireless web interface and RTDE communication of the ESP32 S3 supervisor.

TECHNICAL ANALYSIS & SYSTEM INTEGRATION

This academic research reference documents crucial design paradigms directly relevant to the development of the EB15 six-degrees-of-freedom robotic manipulator. Under the distributed control framework designed for this thesis (featuring the ESP32 S3 as master and the Arduino Uno as slave), the mathematical formulations, communication latencies, and performance profiles analyzed by Quan Leng provide a benchmark for physical and algorithmic modeling. Specifically, this reference establishes solid validation criteria for timing constraints, closed-loop feedback via absolute magnetic encoders (AS5600), and vibration damping.